

Right to Repair in the United States: Definition and Defense Through Strategic Litigation

Afnaan Syed

Clark University

“THE BEST WAY FORWARD FOR THE RIGHT TO REPAIR MOVEMENT IN THE US IS TO FOCUS ON STRATEGIC LITIGATION.”

Abstract

In response to mounting anti-repair practices by major technology manufacturers, recent decades have seen the emergence of a global right to repair movement. It brings together tinkerers, hobbyists, repair professionals, policy advocates, environmental activists, and everyday consumers. This coalition operates both locally and globally to share parts, tools, information, and repair techniques (especially through online forums and Fixit Clinics) and, most importantly, to advocate for legislation protecting the right to repair in all jurisdictions in which it is found. This article examines the right to repair movement with an eye toward achieving three main goals. The first is to establish and defend the basis for the concept of a fundamental right to repair, in both normative and legal contexts – with the latter focusing primarily on the existence of a right to repair in the United States. The second is to highlight how tech corporations in recent decades have taken advantage of copyright law and technological advancement to restrict and monopolize repair markets, and more importantly, how the right to repair movement has fought back on two separate fronts – legislative advocacy and litigation. The third and most important goal of the article is to advance the argument that the best way forward for the right to repair movement in the US is to focus on strategic litigation rather than legislative advocacy by using existing precedent in intellectual property law, namely the longstanding doctrine of patent exhaustion.

Introduction

In response to mounting anti-repair practices by major technology manufacturers, recent decades have seen the emergence of a global right to repair movement. It brings together tinkerers, hobbyists, repair professionals, policy advocates, environmental activists, and everyday consumers. This coalition operates both locally and globally to share parts, tools, information, and repair techniques (especially through online forums and Fixit Clinics) and, most importantly, to advocate for legislation protecting the right to repair in all jurisdictions in which it is found. This article examines the right to repair movement with an eye toward achieving three main goals. The first is to establish and defend the basis for the concept of a fundamental right to repair, in both normative and legal contexts – with the latter focusing primarily on the existence of a right to repair in the United States. The second is to highlight how tech corporations in recent decades have taken advantage of copyright law and technological advancement to restrict and monopolize repair markets, and more importantly, how the right to repair movement has fought back on two separate fronts – legislative advocacy and litigation. The third and most important goal of the article is to advance the argument that the best way forward for the right to repair movement in the US is to focus on strategic litigation rather than legislative advocacy by using existing precedent in intellectual property law, namely the longstanding doctrine of patent exhaustion.

Part 1: Defining the Right to Repair

1.1 - A Normative Basis for the Right to Repair.

Before discussing the legal basis for a right to repair in the United States, the normative question must be addressed – *should* we have such a right? Is an equitable and sustainable society one in which there is a right to repair? On this question, scholars are largely in agreement that the answer is yes, for several reasons. First and foremost are the increasingly apparent environmental consequences of the “replace, don’t repair” culture brought about by repair restrictions. Electronic waste or “e-waste” accounts for roughly 70% of the toxic waste in US

landfills – a figure that continues to rise. That e-waste includes lithium, mercury, and lead—chemicals that endanger our water supplies and threaten human health. [1] Even before the end of the product lifecycle, manufacturing and global shipping produce significant pollution. And extracting the raw materials needed to produce our devices inflicts both environmental and human costs. Take cobalt, for example – a crucial component in the lithium-ion batteries found in phones, laptops, electric vehicles, and countless other consumer electronics. Global demand for cobalt exceeds 100,000 tons per year - a number that is expected to increase threefold over the next decade. [2] Most of the world's cobalt supply is found in the Democratic Republic of Congo. Roughly 20% of it is mined by hand by a quarter million local laborers who work in narrow, unstable tunnels, inhaling toxic cobalt dust, for less than a dollar a day - and an estimated 35,000 of whom are children as young as six years old. [3] By encouraging us to replace rather than repair our devices, manufacturers are increasing the already high demand for materials like cobalt. In the process, they are amplifying the negative environmental and human rights externalities created by global electronics production – and the problem gets worse upon realization that roughly 80% of e-waste globally is not collected for recycling [4], left instead to survive for thousands of years in landfills.

The societal consequences of restrictions on repair are made perhaps more dire when considered in the context of devices used in critical sectors of the globalized economy – for example, healthcare. At the height of the COVID-19 pandemic, as hospitals across the globe faced shortages of life-saving ventilators, and manufacturers scrambled to ramp up production, the ability to maintain and repair existing equipment emerged as a pressing problem. Authorized repair, which often requires shipping devices back to the manufacturer, can leave hospitals without critical equipment for days or weeks. In other instances, manufacturers failed to supply replacement parts, imperiling patients. When a hospital in Chiari, Italy, couldn't secure valves for its respirators from the manufacturer, local volunteers designed and 3D-printed 100 replacements that cost no more than a few dollars apiece. The volunteers managed this feat in just two days, with no help from the manufacturer, which refused to share design specifications. [5]

There is something to be said also about the implications of repair monopolies for exacerbating economic inequality. As Leah Chan Grinvald and Ofer Tur-Sinai argue: “A lack of access to repair information sustains classism because only the wealthy can afford to routinely purchase new products when theirs break. Combined with some manufacturers’ ‘planned obsolescence’ programs, some products are designed to break down sooner than others. For example, Apple has published the lifecycle expectancies of its iPhones, which is approximately three years.” [6] Put simply, making it more difficult to repair products has the effect of sustaining the economic disenfranchisement of those with fewer resources. [7]

Yet aside from staving off worsening economic inequality and the potentially disastrous consequences of repair restrictions on crucial electronic devices that our collective livelihood depends on, scholars in the field argue also that repair’s inherent benefit to an intellectually engaged society is also grounds for its protection. As Aaron Perzanowski has suggested, *The Right to Repair*: “Repair is a social practice that builds valuable skills. It demands analytic reasoning, strategic thinking, and creativity. And repair helps us develop a more complete understanding of how a device operates, enriching our awareness of the world around us. In that sense, it makes us freer, more autonomous, more in control of the world we occupy. Repair empowers us.” [8]

1.2 – The Right to Repair in the US

At the outset, there is an important distinction to be made regarding the term “right to repair.” In current popular discourse, “right to repair” is essentially a blanket term referring to actors on the legislative and legal fronts who advocate for increased access to repair for consumers. This includes fighting back against repair monopolies and pushing for manufacturers to design products that are inherently more repair friendly. This article, however, considers a more literal definition of the words: Just as we in the United States have a right to say whatever we please (with limited exceptions, of course) via the First Amendment, we also have a right to repair the products we own. Recognition of this right has limited relevance in terms of coercing tech manufacturers to make

their products more repair friendly, or to make repair data more accessible. It is, however, crucial for asserting our right to repair the things we own without fear of violating intellectual property law.

Although it is generally accepted among scholars that there is a strong pragmatic basis for the right to repair, there is some disagreement in terms of whether we have this right in the absence of explicit statutory protection.

Some scholars would argue that this right can already be found in the penumbras of existing copyright law. This may seem like judicial overreach at first glance, but the Supreme Court's watershed decision in *Quanta Computer v. LG Electronics* (2008) provides strong precedent suggesting that legitimate ownership of a patented technological device carries with it a degree of control over the use of that product, implicit in which is the right to modify or repair it as one sees fit. The decision does this by relying on a longstanding doctrine in intellectual property law known as the principle of exhaustion. We will proceed with an examination of the origins and theoretical underpinnings of exhaustion, followed by a discussion on the implications of the *Quanta* decision.

Article I, Section 8, Clause 8 of the United States Constitution grants Congress the power " [t]o promote the Progress of Science and useful Arts, by securing for limited Times to Authors and Inventors the exclusive Right to their respective Writings and Discoveries." These exclusive rights often take the form of patents for technological inventions and design processes, e.g. consumer electronics. [9] This provision of the Constitution recognizes that new ideas, once revealed, are easily appropriated by others, and prospective inventors might therefore decline to invest their time, energy, and resources in research without a mechanism to capture some benefit from their inventions. The patent system represents a bargain – society temporarily forgoes the benefits of market competition in return for the public good provided by the inventor's efforts. Thus, patents offer the right to exclude others from making, using, offering for sale, or selling the invention [10] – for a limited period. This period is limited by the exhaustion doctrine, which

declares that once the first sale of a patented item has occurred, the patent holder's rights over that item as set forth by the issued patent are "exhausted." [11] Put simply, exhaustion in its broadest form is what allows us to lend or resell physical copies of books, movies, and other forms of media, as well as electronic devices, at whatever price, if for a price at all, we choose. This might mean an individual lending or reselling to a friend, or it might mean an established bookstore or electronics retailer. However, a thorough examination of the relevant jurisprudence in the United States reveals that patent exhaustion does not merely apply to the reselling rights of the original purchaser. Rather, the courts have held that it applies to use of the patented product as well, as was affirmed in the *Quanta* decision (More on this below).

Exhaustion finds its roots in the common law's historic disfavor of the alienation of personal property. [12] Patent exhaustion as a legal principle in the United States has been expressly recognized by the Supreme Court as early as 1852 in its decision in *Bloomer v. McQuewan*. Here the Court held that when a patented machine passes to the hands of the purchaser from the holder of the patent right, it is no longer within the limits of the monopoly imposed by the patent. [13] 21 years later in *Adams v. Burke*, the Court provided a more clear and broadly encompassing articulation of what patent exhaustion entails: "...in the essential nature of things, when the patentee, or the person having his rights, sells a machine or instrument whose sole value is in its use, he receives the consideration for its use and he parts with the right to restrict that use." [14] Thereafter, the doctrine became fixed in US patent jurisprudence as a kind of axiomatic truth, "more slogan than policy," dutifully repeated but rarely explained. [15] Thus, exhaustion's theoretical underpinnings remain somewhat hazy. But an examination of the case law reveals two primary motivations for the exhaustion doctrine: (1) a desire to discourage unwarranted patentee compensation, and (2) deep misgivings about attaching permanent restrictions to personal property. [16]

As mentioned previously, *Quanta* is perhaps the most significant case for asserting patent exhaustion as the basis for the right to repair. At issue in *Quanta* was a post-sale use restriction that patent owner LG

Electronics imposed on Intel who used the patented processes to install Intel's microprocessors and other components in electronic products. Quanta was an electronic product manufacturer that installed Intel's microprocessors that included LG's patents. Rejecting LGE's claims of patent infringement against Quanta, the Court unanimously affirmed the application of the exhaustion to the use of a patented item after sale, holding: "The longstanding doctrine of patent exhaustion provides that the initial authorized sale of a patented item terminates all patent rights to that item." [17] The Court affirmed the earlier holding in *Bloomer* that any term extensions or other modifications to the patent in question did not affect the rights secured by purchasers who bought the item for use "in the ordinary pursuits of life." [18] Important to note is that the Court also affirmed that, despite LGE's arguments otherwise, exhaustion also applies to method patents – that is, patents granted for processes such as a manufacturing procedure or a recipe for an item, rather than the item itself. Relying on several earlier exhaustion decisions such as *Ethyl Gasoline Corp. v. United States* (holding that the sale of a motor fuel produced under patent also exhausted the patent for a method of using the fuel in combustion motors[19]), the *Quanta* Court held: "...this Court has repeatedly held that method patents were exhausted by the sale of an item that embodied the method...[e]liminating exhaustion for method patents would seriously undermine the exhaustion doctrine. Patentees seeking to avoid patent exhaustion could simply draft their patent claims to describe a method rather than an apparatus." [20] Patent exhaustion, in its most recently articulated form as embodied in the *Quanta* decision, provides as follows: when an individual purchases an item that embodies one or more patents within it, e.g. a desktop computer, the manufacturer of that desktop computer relinquishes its rights to any and all instances of the distribution and use of that product. The individual can face no legal repercussions for lending, reselling, or modifying that instance of the patented product, and that includes practicing the manufacturing methods essential to its function.

It should not come as a surprise that patent exhaustion, especially as embodied in *Quanta*, has been asserted to posit that there is, in fact, a right to repair. As Shubha Ghosh argues, the *Quanta* Court's definition of

“reuse” applies to repair as well: “The right to repair is an example of permitted reuse. At the outset, the right of repair is distinct from reconstructing a product from broken parts, which is an act of unpermitted making of a patented product, rather than permitted reuse.” [21] In doing so, he points to the distinction the Court drew between repair and reconstruction in its 1852 decision in *Wilson v. Simpson* – holding: “ [I]t does not follow, when one of the elements of the combination has become so much worn as to be inoperative, or has been broke, that the machine no longer exists, for restoration to its original use, by the owner who has bought its use. When the wearing or injury is partial, then repair is restoration, and not reconstruction.” [22]

However, it should be clarified that the Court’s distinction here is not infallible. Lower court decisions have affirmed the idea that the difference between repair and reconstruction is a difficult question that must be resolved case by case. [23] The line has been blurry going back thousands of years to the Greek philosopher Plutarch’s famous thought experiment surrounding the ship of Theseus. [24] Federal Circuit Judge Pauline Newman, presiding over the case of *Mallinckrodt, Inc. v. Medipart, Inc.* (1992), conceded that “it is not always clear where the boundary lies: How much ‘repair’ is fair before the device is deemed reconstructed.” [25] Mark Janis, in his seminal survey of the repair versus reconstruction jurisprudence in 1999, observed: “Courts long ago abandoned all efforts to cabin the repair-reconstruction dichotomy within a rigid framework of rules. Instead, they rest their decisions on ‘the exercise of sound common sense and an intelligent judgment.’” [26] According to Janis, the current test is based on “spentness”: Refurbishment is permissible as long as the product still retains some useful life. Janis argues that this test is indeterminate and better off replaced with a test based on the intention of the parties. [27]

In addition, some scholars, most notably Grinvald and Tur-Sinai, ground the right to repair in the utilitarian notion of intellectual property rights as a governmental tool to bargain for the development and disclosure of socially valuable information. [28]

Other scholars, however, seem to believe notions of a penumbral right

to repair within US patent exhaustion jurisprudence is insufficient by itself, and argue that codifying the right to repair in legislation is necessary to protect this right going forward, particularly in the face of new repair restrictions brought about by a clever combination of technological innovation and copyright law that tech corporations have leveraged. [29] In other words, the repair rights encompassed within patent exhaustion have been eroded by advancements in technology and instruments of copyright and contract law. In the words of Nicholas Mirr: "...the existing framework of copyright law and enforcement in the United States is insufficient to protect consumers' right to do what they will with their products. Through End User License Agreements and other contracts, companies are effectively able to limit the right to repair via contract even if there were to be a drastic overhaul of our current copyright system." [30] Mirr then goes on to argue that Congress should adopt legislation requiring companies to facilitate the repair process and should incorporate a "degree of repairability" requirement in the legislation to ensure the ability to repair remains an option for most consumers. [31] In other words, scholars such as Mirr argue that there at the very least should be a right to repair, but it must be codified through legislation, ideally at the federal level.

There are, therefore, at least two clear domains - litigation and lawmaking - in which the right to repair movement is playing out. The primary aim of this article is to evaluate which of these domains offers the more fertile ground for the newly emergent social movement surrounding the right to repair to mobilize. As the analysis below demonstrates, of these two policymaking venues, mobilization through strategic litigation is likely to be the more effective of the two. In other words, the issue of the right to repair should be "venue shopped" from the legislature to the courts. For this argument to work, it is a necessary assumption to make that there is, in fact, a right to repair that has been implicitly acknowledged by the courts, most notably the Supreme Court in its recent patent exhaustion jurisprudence as embodied in *Quanta* and other recent cases. This may be a tall order, considering the clever use of copyright law and technological innovation that tech giants have leveraged to monopolize repair in recent years. However, recall the previous discussion about the theoretical underpinnings of

patent/copyright exhaustion pertaining to the common law's disfavor of placing external restraints on one's personal property, along with the fact that the concept of exhaustion itself does not appear anywhere in the text of the US Constitution, and yet has been acknowledged and reiterated repeatedly by the Supreme Court. These factors, along with more recent lower court decisions in actions brought in the name of fighting back against repair restrictions (to be discussed in Part III) make clear that there is, in fact, a right to repair in the US that can be protected through litigation, even if not yet explicitly acknowledged by a court of law.

I proceed below with a discussion of how a social movement surrounding a right to repair came to emerge in the US. Then, I describe how corporations have launched a coordinated and sometimes quite effective campaign against the movement. This is followed by a discussion of how repair activists have fought back against corporations. These efforts have ranged from lobbying for legislation to antitrust litigation, each with somewhat mixed results. In Part 3, I evaluate the accomplishments of the movement in these two domains and suggest that strategic litigation is likely to be the most effective strategy for activists and others in the movement for a right to repair to succeed. Lawmaking depends on the vicissitudes of public attitudes and the political mood of the country. The legal case for a right to repair is much more stable because such a right can be independently justified drawing upon our common law tradition, doctrine, and case law.

Section II: Attack on Repair

The necessity of the growing right to repair movement across the globe is owed to the fact that technology corporations, in recent years, have adopted increasing measures to prevent consumers from repairing the products they own either by themselves or through an IRP (independent repair provider). This is done in the name of monopolizing repair markets and ultimately boosting profits by encouraging us, when our devices break, to replace our devices rather than repair them. [32] These measures take many forms – intellectual property law, hardware design, software restrictions, pricing strategies, and marketing messages – but for the purposes of this study, it is viable and somewhat

necessary to consolidate them into two main categories: anti-repair technological design and copyright law. Let's examine these one at a time.

2.1 - Anti-Repair Technological Design

There was a time when a clear line could be drawn between products and services. Once you purchased a TV, for example, the manufacturer had little say over how you used it. This is clearly different than purchasing a subscription to a streaming service to use on the TV you bought, such as Netflix or Disney+. In the latter case the service provider can add or subtract from the selection of the movies and shows your subscription grants you access to, change the price, or cancel the service altogether. Inconvenient as this may be, it is a reasonable thing to expect when purchasing access to an exclusively software-based streaming service that is accessed through the "cloud" – a distributed network of servers hosted in external data centers all over the world. [33] Today, however, the line between product and service is more of a blurry smudge. Our TVs, cars, appliances, smartphones, and countless other technological products we use every day are bundled with deeply intertwined software and external data services that are central to their functionality – a phenomenon that British technologist Kevin Ashton in 1999 aptly named "The Internet of Things." [34] The implications for repair are apparent and ominous.

2.1.1 - Apple

In 2016, several iPhone users reported that their devices had "bricked" - in other words, the phone had become completely unusable and thus turned into an expensive brick - after replacing a screen or Touch ID home button themselves or at another location not authorized by Apple. It was revealed that "Error 53" was a security measure implemented by Apple to prevent the devices from operating properly if they detected that third party components had been installed. Apple claimed this error was caused by a security check designed to prevent the Touch ID fingerprint reader from being exploited. Eventually, after complaints piled up, Apple released a software update which eliminated the error and restored the owner's phone to working condition. As it turned out, Australian courts declared Error 53 a violation of Australian consumer law, and Apple was eventually fined AUD \$9 million in June of 2018 as a result. [35]

2.1.2 - HP

“Security measures” like Error 53 are unfortunately not limited to cell phones. In March of 2025, tech giant Hewlett Packard (more commonly known as HP) settled a class action lawsuit in which it was accused of unlawfully blocking customers from using third-party toner cartridges. One of the named plaintiffs in the case, Mobile Emergency Housing Corp, an organization that works with emergency management organizations and government agencies to provide shelters for disaster victims and first responders across the US and Caribbean, bought an HP Color LaserJet Pro M254 in August 2019. In October 2020, the organization used toner cartridges from third-party supplier Greensky rather than pay for HP’s premium-priced toner. A month later, HP sent or activated a firmware update - part of its so-called “Dynamic Security” measures - rendering MEHC’s printer incompatible with third-party toner cartridges like those from Greensky. When MEHC CEO Joseph James tried to print out a document from the printer, he got a “Supply Problem” error message that rendered the printer completely unusable. [36]

2.1.3 – John Deere

If anyone could escape the repair restrictions brought about by the Internet of Things, you might think it would be farmers. We often imagine farmers living off land in rural communities, less dependent on modern technology and more rooted in traditions and practices that predate the digital era. But this image of farm life is out of step with reality. Farming today relies on a range of technologies, from moisture sensors and drones to genetically modified crops and patented biological pest controls. Even that classic symbol of rural America, the John Deere tractor, has been transformed into complex, software dependent piece of digital technology. [37]

Modern John Deere tractors can cost as much as \$800,000. No longer purely mechanical devices, they depend on multiple electronic control units (ECUs) to operate everything from the engine to the power seat. These embedded computers run software code essential to the operation and repair of the tractor. By controlling access to that code, John Deere can prevent independent diagnosis and repair. Without

enlisting a John Deere technician, the tractor's software won't even recognize replacement parts. That level of control forces farmers to rely on authorized John Deere dealers for service, rather than doing it themselves or turning to local IRPs. [38] So, a farmer who buys something as simple as a turn signal and installs it themselves or pays an independent repair shop to install it is still forced to pay the local John Deere dealer to send a technician to authenticate the part before it will work, for a fee of \$230, plus \$130 an hour for the technician. [39]

2.2 – Copyright Law

Aside from hostile design, one of the key legal tools firms have relied on to restrict repair is § 1201 of the Digital Millennium Copyright Act (“DMCA”). That statute makes it unlawful to circumvent - that is, to remove, bypass, or disable - technological protection measures (“TPMs”) that restrict access to copyrighted works. In addition, § 1201's antitrafficking provisions prohibit the creation, sale, or distribution of tools that enable circumvention. [40] Device makers routinely use TPMs – such as the previously mentioned ECUs - to limit access to the software code that controls devices from smartphones to cars. That code is often necessary to diagnose and repair devices – see the previous discussion on the ECUs in John Deere tractors. These digital locks already pose practical hurdles for repair. But § 1201 compounds those difficulties by introducing potential criminal liability for repairers who remove or bypass such digital locks. [41]

The DMCA was enacted in 1998 to update and modernize US copyright law in response to the challenges presented by digital technologies and the internet. [42] At the time of the DMCA's inception, that largely meant protecting the ownership rights of producers of digital media from unauthorized access to reproduce their works through decryption software. But as TPMs and Digital Rights Management (“DRM”) features have become increasingly intertwined into consumer technologies, the statute has been effectively hijacked by technology manufacturers to restrict consumers' right to repair. The copyright protection afforded to the software code that is necessary to diagnose and repair our devices has, at least for now, overridden the patent exhaustion doctrine that theoretically protects this right. With that in mind, it should not come as

a surprise that the oppressive power afforded to technology giants by the DMCA has been challenged in courts of law, as will be discussed in further detail below.

Section III: The Right to Repair Movement

In response to these anti-repair practices, recent decades have seen the emergence of a global right to repair movement. It brings together tinkerers, hobbyists, repair professionals, policy advocates, environmental activists, and everyday consumers. [38] This coalition operates both locally and globally to share parts, tools, information, and repair techniques (especially through online forums and Fixit Clinics[39]) and, most importantly, to advocate for legislation protecting the right to repair in all jurisdictions in which it is found.

3.1 - Fight for Repair in the Legislature

Although there has been a strong push for right to repair legislation in the United States, the path has been made complicated by lobbying from tech companies. The rest of this section will briefly examine the history of attempts to codify the right to repair in various forms through legislation in the US.

In 2012, Massachusetts became the first state to take action preserving the right to repair when it passed the Act Protecting Motor Vehicle Owners and Small Businesses in Repairing Motor Vehicles. [45] Beyond its implications for vehicle repair, the Massachusetts automotive repair law has served as the template for a broader, nationwide effort to enshrine consumers' right to repair in state law [46] that has intensified in the past few years. In 2021, a total of twenty-seven states considered right to repair legislation, twelve of which were aimed specifically at agriculture equipment. In 2023, that number rose to thirty-three. Federal repair bills were also introduced in 2022 and 2023 in the US Senate and the House of Representatives respectively. [47] And as of February 2025, all 50 states have introduced right to repair legislation. [48] These bills have attempted to protect the repair rights of both consumers and IRPs in various ways. Some are narrowly tailored to protect access to repair information specifically for farming equipment,

in response to the more ubiquitous repair restriction practices by John Deere. [49] Others are more broad, such as Iowa's H.F. 556 introduced in March of 2017, which covers any and all "digital electronic products." [50] Finally, the Right to Repair Bill introduced in Washington in 2018 presents an approach that went a step further - it would prevent the sale of devices in that state which are not easily able to be repaired by independent repair providers. Washington's focus on legislating certain levels of reparability made it more comprehensive than any other bill yet proposed. [51]

Introducing legislation is one matter. Getting it passed in the face of the often prohibitively lengthy and contentious legislative process is another entirely – and in the case of repair, the problem has been exacerbated by tech corporations waging a fierce lobbying campaign against proposed laws protecting the right to repair. In Nebraska, for example, the 2017 Fair Repair Act was met with staunch opposition by tech companies like Apple and AT&T, as well as from John Deere and other agriculture-focused tech manufacturers and trade organizations. [52] This level of opposition is representative of the challenges facing the Right to Repair movement nationwide.

Despite this, the first half of the 2020s have seen some notable state-level successes. In 2020, Massachusetts voters were able to close a loophole in the 2012 statute protecting the right to repair for automobiles by voting "yes" on a ballot measure that would require car makers to share information they collect wirelessly about vehicles' mechanical health. Such information, known as telematics, can help anticipate problems before they arise. [53] Elsewhere, Washington was finally able to sign right to repair protections for consumers into law in the form of two bills: H.B. 1483 and S.B. 5680, both of which were signed into law in May of 2025. [54]

That said, the anti-repair coalition formed by technology giants remains a substantial barrier to legislative efforts to codify the right to repair for consumers and IRPs in both state and federal law, which brings us to the other major stage of the fight for repair: the courts.

3.2 – Fight for Repair in the Courts

Alongside the struggle to codify the right to repair in legislation, some consumer groups, in conjunction with the federal government, have taken it upon themselves to protect the right to repair through the judicial process. Before I illustrate some of these proceedings, here is a caveat: Much of the litigation undertaken in the name of protecting repair rights is still ongoing and, at the time of this writing, there is very little in the way of precedential court decisions from which inferences can be drawn. But what we can do is examine some of the most salient ongoing cases in this realm and what they might mean for the fight for repair in the courts going forward.

3.2.1 - Alliance for Automotive Innovation v. Campbell

The passage of the updated right to repair law for automobiles in Massachusetts in 2020 was followed by a slew of legal challenges by auto manufacturers, the last of which were dismissed in February of 2025, allowing the law to take effect. These challenges were brought through the Alliance for Automotive Innovation (“AAI”), a lobbying group representing more than forty major automakers and other tech manufacturers such as Ford, Mazda, Toyota, Mercedes, and many more. [55] AAI filed suit against Massachusetts Attorney General Andrea Joy Campbell, arguing that the law requiring the disclosure of telematic data was preempted by existing federal car safety laws – specifically, the Motor Vehicle Safety Act (“MVSA”) and several federal motor vehicle safety standards promulgated by the National Highway Traffic Safety Administration. This challenge failed, and the law was upheld. [56]

Despite being primarily concerned with questions of federal preemption and not directly addressing the implications of these questions on the existence or recognition of a “right to repair”, The District Court’s decision in *AAI v. Campbell* as the first major ruling in a slew of recent repair litigation should be interpreted as an instructive victory in the fight for repair in the courts. Regardless of what their actual motives may be for pushing back against the right to repair movement, tech corporations argue first and foremost that removing the repair restrictions they have levied on their devices through TPMs and embedded software poses risks for safety and data security. The AAI

decision sets the major precedent that courts disagree with this argument, and that state-level right to repair laws are workable in conjunction with existing federal regulations on the devices we own.

3.2.2 - Ongoing Deere Antitrust Litigation: Forest River Farms and the FTC

In January of 2022, multiple suits filed by farms across the country against John Deere were consolidated into a class action complaint in the US District Court for the Northern District of Illinois (Deere's home state) filed as *Forest River Farms v. Deere & Co.* The suit alleged, first and foremost, that Deere's repair restrictions on its farming equipment (as discussed in 3.1.1.3) violate the Sherman Antitrust Act "by monopolizing or attempting to monopolize the Deere Repair Services Market in a manner that harmed competition and injured the purchasers of such services by reducing choice and increasing prices in this market to supracompetitive levels." [57] The farms alleged also that Deere has leveraged its monopoly power over Deere software to tie sales of its tractors to sales of Deere Repair Services, in violation of Section 2 of the Sherman Act, and the farming giant has unjustly enriched itself by profiting from farmers' payment of supracompetitive prices for Deere Repair Services in violation of the antitrust laws. Alongside declaratory and injunctive relief, as well as treble and exemplary damages, the plaintiff farmers sought an order requiring Deere to make the necessary software available, at reasonable cost, to individuals and repair shops. [58] Three years later in January of 2025, the Federal Trade Commission filed its own suit, along with the states of Minnesota and Illinois, against Deere in the same district court, alleging much of the same. [59]

In both the 2022 class action suit led by Forest River Farms and the FTC action in 2025, Deere attempted to have the cases thrown out by the District Court. Motions to dismiss were filed by Deere's counsel in both cases and were each promptly denied by District Judge Iain Johnston. [55] Both cases are ongoing as of now, and no additional major ruling has been made in either. Regardless, this litigation still warrants mentioning because it shows how turning to litigation to fight back against the attack on repair, even if in the context of antitrust law rather

than protecting a conceptual “right to repair”, is a novel strategy by consumers and regulators that is rapidly gaining traction – and as seen from the AAI decision and the failed motions to dismiss by Deere, the results seem promising. Zooming out from its action against Deere in the District Court, the FTC along with the Department of Justice has also made ground in pushing back against the oppressive power of the DMCA in restricting repair.

3.3 – DMCA Repair Exemptions

As mentioned previously, copyright law – specifically, § 1201 of the DMCA – has been the critical legal instrument levied by tech manufacturers to restrict repair. This is because that provision normally prohibits bypassing technological protection measures (TPMs), even for lawful purposes like repair. However, the US Copyright Office’s triennial rulemaking process creates temporary exemptions (renewed every three years) to balance copyright protection with consumer rights. [61] Recent years have seen an increasing number of exemptions granted by the Copyright Office to § 1201 allowing circumvention of TPMs for repair purposes in the name of protecting repair. The first series of repair exemptions was granted in 2015 and allowed consumers and IRPs to circumvent the digital locks on motorized land vehicles. [62] Since then, the Copyright Office has expanded its repair exemptions to include smartphones and tablets, medical devices, industrial and some agricultural equipment, and commercial food preparation equipment as of the latest rulemaking process in 2024. [63] These exemptions have been advocated by the FTC and the DOJ, who in 2024 filed a joint comment with the Copyright Office supporting the renewal and expansion of repair exemptions. The joint comment argued that by limiting access to the data and software needed for independent repair and maintenance, TPMs can be used to squash competition for replacement parts, repair, and maintenance, which ultimately limits consumers’ and businesses’ choices and raises costs. [64]

Because the Copyright Office is technically a unit of the legislative branch, operating as a service unit of the Library of Congress, some may argue that the DMCA exemptions should be categorized under the legislative front of the fight for repair. I argue, however, that the

exemptions reflect not necessarily a change in the law but an updated interpretation of it, and moreover the two main government agencies argue that the DMCA exemptions should be categorized under the legislative front of the fight for repair. I argue, however, that the exemptions reflect not necessarily a change in the law but an updated interpretation of it, and moreover the two main government agencies advocating for the exemptions (the FTC and DOJ) operate as law enforcement agencies and often deal with litigation, therefore these exemptions fall more neatly under the litigation front. Regardless, the recent repair exemptions to § 1201 of the DMCA granted by the Copyright Office are a crucial victory for the right to repair movement, as advocates and regulators have been able to significantly weaken the primary legal bottleneck for protecting repair rights. But these exemptions are temporary and limited in nature, and thus ultimately represent only a temporary solution to the issue of the right to repair. With this in mind, let's move onto discussing the most effective strategy for protecting the right to repair going forward.

Section IV: Repair Going Forward: Strategic Litigation

Earlier I stated that it is a necessary assumption to make for purposes of writing this article that there does exist a “right to repair” embodied within US patent exhaustion jurisprudence. I would argue that this right exists, and the nomenclature of the “right to repair” movement in recent decades suggests that proponents of the movement would agree. But the only way to have this right explicitly acknowledged by the courts is through strategic litigation through a test case that incorporates the proper requirements for patent exhaustion but also represents a violation of the DMCA. Let's examine these factors one at a time.

4.1 - Strategic Litigation as a Force for Change

The power of strategic or “impact” litigation as a force for change in American society is too well-demonstrated historically to be denied. Some of our most critical civil rights that we enjoy today are the result of strategic litigation. These include the outlawing of racial discrimination through *Brown v. Board of Education* (1954), which was the culmination

of a strategic litigation campaign by the NAACP [65]; the constitutional right to privacy through *Griswold v. Connecticut* (1965), in which C. Lee Buxton of the Yale School of Medicine and Estelle Griswold of Planned Parenthood in Connecticut opened a birth control clinic in New Haven solely for the purpose of using the clinic to challenge the constitutionality of Connecticut's anti-contraceptive statute[66]; and the constitutional right to same-sex marriage through *Obergefell v. Hodges* (2015), in which groups of same-sex couples sued their relevant state agencies in Ohio, Michigan, Kentucky, and Tennessee to challenge the constitutionality of those states' bans on same-sex marriage or refusal to recognize legal same-sex marriages that occurred in jurisdictions that provided for such marriages. [67]

What these cases all have in common - and that the recent litigation efforts undertaken in the name of fighting back against repair restrictions thus far have lacked - is that they utilized grievances through test cases that sought to affect broad societal change. There is certainly something to be said about the risk of relying on strategic litigation in terms of short-circuiting the democratic process – that is, relying on a unilateral court order rather than elected representatives to protect our civil rights. But that is not a major concern of this article. The fact of the matter is that if the goal of the right to repair movement is indeed to protect a “right to repair”, then temporary exemptions to an oppressive statute can only go so far, and strategic litigation presents a promising opportunity to protect this right in a more meaningful and permanent way. The final section will discuss why and how this is the case.

4.2 – Patent Exhaustion vs. the DMCA

By all accounts, the principle of patent exhaustion on its face, and especially in its latest interpretation by the Supreme Court in the *Quanta* decision, protects the right to repair in most scenarios – see 1.2. The issue in recent years is that, as is often the case, technological innovation has outpaced the law, and tech manufacturers have made an end run around patent exhaustion by claiming copyright protection through the DMCA over the embedded software code that is increasingly essential to diagnose and repair the devices we use on a day-to-day basis. The patent on the device itself may be exhausted, but the

software essential to its function remains under the technological and legal control of the manufacturer. It is true that the copyright protection afforded to tech manufacturers by § 1201 of the DMCA has effectively overridden patent exhaustion, but this is the case only because no litigation has occurred yet that would force the courts to answer this question conclusively. When we consider the numerous exemptions that have been granted to § 1201 by the Copyright Office for repair purposes as covered in 2.4, as well as the general apparent sympathy of the courts to the right to repair movement thus far as covered in 2.3, it seems likely that if this question is eventually brought before a court, the ruling would bolster patent exhaustion and restore it to its former preeminence in dictating what manufacturers may or may not control over how we use their products. This is especially true when we consider what the Supreme Court said about method patents specifically in the *Quanta* decision. In the words of Justice Thomas, writing for a unanimous court: "It is true that a patented method may not be sold in the same way as an article or device, but methods nonetheless may be 'embodied' in a product, the sale of which exhausts patent rights." [68] Applying this reasoning to TPMs, the questions arise: Would the software code essential to the function, diagnosis and repair of a car, tractor, or smartphone not exemplify a method for that device? And since that code is quite literally embedded into the device itself, would any patent or copyright protection on the code not also be exhausted by that device's sale?

There is something to be said also about what the numerous repair exemptions granted to the DMCA say about the robustness of that statute. As stated previously, the DMCA was enacted almost thirty years ago, and the original intent of the statute – especially § 1201 – was mainly protecting the ownership rights of producers of digital media from unauthorized access to reproduce their works through decryption software. That tech firms have been able to levy the power of the statute to restrict third-party repair through TPMs is less an intentional goal and more an unforeseen implication. Perhaps this is why the Copyright Office has granted so many exemptions to it. But regardless, the question arises: if the Copyright Office is willing to grant so many exemptions to § 1201 of the DMCA, then what reason is there to not

rewrite or overturn the provision? My argument on this front is that this provision should be drastically reformed to eliminate the need for constantly having to grant and renew numerous exemptions for repair purposes for specific types of devices, or, if pursued through litigation, then it should be struck down entirely on grounds of violating the patent exhaustion principle and thus being unconstitutional as applied.

The right to repair movement thus far, both on the legislative and litigation fronts, has been rather decentralized. The initiative has attracted a diverse yet scattered coalition of sympathetic lawmakers, environmental activists, repair-industry organizations such as iFixit, and more general consumer-focused advocacy groups such as PIRG. [69] The drawback to this grassroots nature is that there is no single organization to consolidate the movement and pursue the most effective strategy going forward, such as what civil rights had in the NAACP. While lawmakers introduce state-level codified repair protections for certain categories of devices, other groups (see the previous discussion on *Forest River Farms v. Deere*) have taken it upon themselves to take legal action pursuant to existing antitrust law – both disjointed and temporary solutions in terms of defending repair as a fundamental right. Pursuing strategic litigation would solve this problem and, based on early rulings of the courts in recent repair restriction litigation as previously discussed, alongside the Supreme Court’s recent patent exhaustion jurisprudence, provides the best possible chance of providing a permanent protection for an actual “right” to repair. This of course would necessitate a much higher level of coordination between right to repair advocates for litigation purposes, especially in terms of utilizing a test case. This test case would have to satisfy the requirements for patent exhaustion but also involve a circumvention of TPMs in violation of § 1201 of the DMCA, with the primary goal being to argue that the DMCA provision is overridden by the doctrine of patent exhaustion.

As with any high-stakes litigation, there is, of course, a real risk of setting adverse precedent. However, in this case that risk is simply not large enough to preclude strategic litigation as a viable strategy. In terms of the previously discussed recent repair litigation, from AAI to

the Deere antitrust suits, all major rulings have come up in favor of repair. Perhaps it could be argued that the chances for strategic litigation pursuant to patent exhaustion are less certain, but this is only because this avenue has yet to be pursued. If recent jurisprudence regarding patent exhaustion in general (see *Quanta*) is anything to go by, then applying this principle to repair – an issue which courts thus far have apparently been sympathetic to – is a straightforward decision to make.

Moreover, as it stands, tech giants have been allowed free rein to use § 1201 of the DMCA as they please to lock down devices and inhibit repair or modification of any kind by the end user for the entirety of the product's life, making an end run around patent exhaustion entirely. As stated previously, this is the case only because the issue has not been brought to court. If strategic litigation of the kind advocated for in this article is undertaken and fails, thereby generating adverse precedent, then repair will be in the same position it is now, and the focus will have to be put solely on lobbying legislatures to update our outdated intellectual property laws and codify repair as a consumer protection. On the other hand, if such litigation were to prevail, either striking down the provision entirely or stripping its protection from TPMs, it would be perhaps the greatest victory for the right to repair movement in the US yet.

The avenue of strategic litigation for the right to repair movement is made more apparent by the nature of technological innovation on its face, and more specifically how it interacts with politics. It is hardly novel to say that technological innovation is outpacing the law. The appropriation of § 1201 of the DMCA by tech giants to claim copyright protection over software code embedded into physical devices is proof of this. While keeping in mind the risk of legislative appropriation, it is in situations like this where impact litigation may serve its most important purpose. The role of the courts, after all, is to interpret the law. A pristine example, although not explicitly strategic litigation, can be found in *In re Search of a Residence in Oakland*. Decided by the US District Court for the Northern District of California in January of 2019, this case dealt with a search and seizure warrant application by federal

authorities for a residence in Oakland, CA. As part of the application, federal authorities also sought the authority to compel any individuals present at the time of the warrant execution to unlock any digital device found via biometric means (i.e. a fingerprint scanner) for that device's contents to be searched. The District Court denied this warrant application on grounds of violating the Fifth Amendment privilege against self-incrimination. [70]

The Court reasoned that although the witness is not technically being compelled to speak, compelling the utilization of biometrics to unlock a device with potentially incriminating information on it is functionally equivalent to compelling a witness to reveal an alphanumeric passcode, thereby rendering it testimonial under the Fifth Amendment. [71] As part of her opinion, Magistrate Judge Kandis Westmore stated the following: "The challenge facing the courts is that technology is outpacing the law. In recognition of this reality, the United States Supreme Court recently instructed courts to adopt rules that 'take account of more sophisticated systems that are already in use or in development.'" [72] The same reasoning should be applied to TPMs and the DMCA.

4.3 – Patent Exhaustion vs. Contract Law (EULAs)

There is one major legal hurdle that has yet to be discussed, and which would present a serious challenge to the right to repair movement in the courts if strategic litigation is pursued – contract law. More specifically, End User License Agreements, commonly referred to as EULAs. An EULA is a type of contract between software publishers and end users, which governs the end user's right to use software, and is thus extremely important as it prescribes what consumers may and may not do with the product. Tech companies have long since adopted the strategy of weaponizing EULAs to explicitly preclude repair or modification of almost any kind by the end user. John Deere, for example, requires that all farmers that purchase their equipment sign an EULA that forbids nearly all repair and modification to farming equipment. [73]

Apple takes it a step further. Anyone who has purchased a MacBook is all too familiar with the experience of booting it up for the first time and

being greeted with a lengthy End User License Agreement. Most are also familiar with the routine of ignoring this EULA and dragging the slider on the screen to the bottom as fast as possible to activate the “Accept Terms and Conditions” button. But take the time to read it through, and you will find that although you have technically purchased the laptop itself, the software powering it has in fact only been licensed to you by Apple for reproduction, modification, publication and distribution of non-copyrighted materials, materials in which you own the copyright. The agreement also expressly forbids unauthorized modifications to the MacBook, such as by disabling hardware or software controls. [74]

The obstacle that EULAs present to the prospect of strategic litigation pursuant to patent exhaustion are apparent. Under this regime, it is not only TPMs but binding contract agreements that restrict the right to repair. How may we claim that a plaintiff’s property rights have been violated when they voluntarily agreed to not repair or modify the device? However, contract law is not the trump card to shut down any invocation of exhaustion that it may appear to be.

This article does not intend to present an exhaustive list of ways that repair advocates may overcome the contract obstacle in court. But upon preliminary review, there is a route to take that is quite simple: The principle of patent exhaustion is robust enough to override restrictive EULAs completely, rendering them null and void for purposes of repair and modification.

This suggestion may seem overly optimistic; perhaps even delusional. But consider the following: John Deere is far and away the pre-eminent manufacturer of commercial agricultural equipment, boasting a 70% market share within the industrial machinery and components industry during 2025. [75] Over in the world of personal computing, Apple may not boast the same market power on its own, but almost every major computer manufacturer, such as Lenovo and HP, now leverage similarly restrictive EULAs on all their products. [76]

In other words, the freedom of contract that may be invoked by tech giants in court to justify repair restrictions is not actually free. While a much more pragmatic interpretation of the law by a court would be

required, the argument can and should be made that end users are not “voluntarily” signing away their right to repair and modify what they own as they please in any real sense of the word.

But even ignoring these circumstances, what is even more crucial to note is that the *Quanta* decision (cited heavily throughout this article and which strategic litigation pursuant to patent exhaustion in the name of repair must also cite) also involved a breach of contract. Stipulated to by all parties to that case is the fact that LGE’s licensing agreement with Intel explicitly stated that the license does not extend to a product made by combining an Intel product with a non-Intel product, and that Intel must give its customers written notice of this. Stipulated to also is the fact that one of Intel’s customers – Quanta Computer, Inc. – disregarded this part of the agreement and manufactured computers using the licensed Intel parts with non-Intel parts. [77] A clear breach of contract was present, and yet the breaching party prevailed under patent exhaustion. The goal of the strategic litigation advocated for in this article is already to prove in a court of law that the exhaustion principle under the common law tradition has enough force to override TPMs and federal copyright law. To prove also that it overrides EULAs is little more than a minor adjustment, and the precedent is there.

Conclusion

Patent exhaustion jurisprudence supports the idea that the concept of an inherent right to repair does indeed exist. But because the grassroots political movement for protecting this right is – relatively speaking – very young, it shouldn’t come as a surprise that in its current form it is rather decentralized and hasn’t yet been able to decide on the most effective strategy to focus all its efforts on, despite the victories it has achieved in the legislature and the courts. That being said, because the ideal outcome for the recently emerged right to repair movement is to in fact gain recognition for an inherent “right” to repair the things we own, and because the courts thus far in proceedings pertaining to repair restrictions have been sympathetic to the movement, coupled with the fact that technological innovation and the DMCA have left open several questions as they pertain to patent exhaustion, strategic litigation is the most effective strategy going forward for protecting the right to repair in the United States.

Notes

- [1] Aaron Perzanowski, "Consumer Perceptions of the Right to Repair," *Indiana Law Journal* 96, no. 2 (May 1, 2020): 363.
- [2] *Id.* at 364.
- [3] *Id.*
- [4] Leah Chan Grinvald and Ofer Tur-Sinai, "Defending the Right to Repair," in *Feminist Cyberlaw*, ed. Meg Leta Jones and Amanda Levendowski (Berkeley: University of California Press, 2024), 30.
- [5] Perzanowski, "Consumer perceptions of the right to repair," 364.
- [6] Grinvald and Tur-Sinai, "Defending the Right to Repair," 29.
- [7] *Id.* at 30.
- [8] Aaron Perzanowski, *The Right to Repair: Reclaiming the Things We Own* (Cambridge: Cambridge University Press, 2022), 10.
- [9] "Patent Essentials," United States Patent and Trademark Office, February 18, 2025, www.uspto.gov/patents/basics/essentials.
- [10] Andrew T. Dufresne, "The Exhaustion Doctrine Revived? Assessing the Scope and Possible Effects of the Supreme Court's *Quanta* Decision," *Berkeley Technology Law Journal* 24, no. 1 (2009): 13.
- [11] Note that this is a separate term from, but closely related to, the "first sale" doctrine in copyright law, which limits a copyright owner's distribution right such that their right to control the sale of a particular copy of a work is terminated once the owner releases that copy to the public through a sale, gift, or loan. Kimberley Byer, "The Death of the First Sale Doctrine," *Journal on Telecommunications & High Technology Law* 11, no. 2 (Fall 2013): 392.
- [12] *Id.* at 392.
- [13] *Bloomer v. McQuewan*, 55 U.S. 539, 549 (1852).
- [14] *Adams v. Burke*, 84 U.S. 453, 456 (1873).

- [15] Dufresne, "The Exhaustion Doctrine Revived?" 12–13.
- [16] *Id.* at 12-13.
- [17] *Quanta Computer, Inc. et al. v. LG Electronics, Inc.*, 553 US 617, 625 (2008).
- [18] *Id.* at 625.
- [19] *Ethyl Gasoline Corp. v. United States*, 309 U.S. 436, 457 (1940).
- [20] *Quanta Computer, Inc.*, 553 U.S. 617, 629 (2008).
- [21] Shubha Ghosh, "The Continuing Right to Repair," *Berkeley Technology Law Journal* 37, no. 3 (2022): 1103.
- [22] *Wilson v. Simpson*, 50 U.S. (9 How.) 109, 123 (1850).
- [23] Liu, Benjamin. "Toward a Patent Exhaustion Regime for Sustainable Toward a Patent Exhaustion Regime for Sustainable Development." *Berkeley Journal of International Law* 32, no. 2 (December 2014).
- [24] Perzanowski, The Right to Repair, 129. "The sequential replacement of parts hints at a deeper puzzle that dates back at least 2,000 years. Plutarch described how the Athenians preserved the ship that carried Theseus home from Crete. 'They took away the old planks as they decayed, putting in new and stronger timber in their places.' Among philosophers, the ship of Theseus embodied the problem of identity over time. Was the preserved ship the same vessel Theseus captained, or was it a new ship altogether?"
- [25] *Mallinckrodt, Inc. v. Medipart, Inc.*, 976 F.2d 700 (Fed. Cir. 1992).
- [26] Janis, Mark D. "A Tale of the Apocryphal Axe: Repair, Reconstruction, and the Implied License in Intellectual Property Law." *Maryland Law Review* 58, no. 2 (1999)
- [27] *Id.*
- [28] Grinvald and Tur-Sinai, "Defending the Right to Repair," 29.
- [29] Perzanowski, *The Right to Repair*, 72-110.

[30] Nicholas A. Mirr, "Defending the Right to Repair: An Argument for Federal Legislation Guaranteeing the Right to Repair," *Iowa Law Review* 105, no. 5 (July 2020): 2393.

[31] *Id.* at 2393-2428.

[32] *Id.* at 10.

[33] "What Is the Cloud?," Cloudflare, accessed November 4, 2025, www.cloudflare.com/learning/cloud/what-is-the-cloud/.

[34] Samuel Greengard, "Internet of Things," *Britannica*, November 14, 2022, www.britannica.com/science/Internet-of-Things.

[35] Mirr, "Defending the Right to Repair", 2397.

[36] Iain Thomson, "HP Settles Lawsuit after Killing First Responder's Printers," *The Register*, March 19, 2025, www.theregister.com/2025/03/19/hp_printer_lawsuit_settled/; see also "Notice of Proposed Class Action Settlement," HP, December 24, 2024, support.hp.com/ie-en/document/ish_11775540-11775594-16.

[37] Perzanowski, *The Right to Repair*, 9.

[38] *Id.*

[39] Perzanowski, "Consumer Perceptions of the Right to Repair", 370.

[40] 17 U.S.C. § 1201 (2018).

[41] Perzanowski, "Consumer Conceptions of the Right to Repair", 370-371.

[42] Grinvald and Tur-Sinai, "Defending the Right to Repair.", 26-27. See also Jane Ginsburg, "Copyright Legislation for the 'Digital Millennium,'" *Columbia VLA Journal of Law & the Arts* 23 (1999): 137-179.

[43] Perzanowski, *The Right to Repair*, 10-11.

[39] *Id.* at 11.

[45] The Massachusetts 2012 bill required manufacturers of motor vehicles to “make available for purchase by owners . . . and by independent repair facilities the same diagnostic and repair information” as well as “all diagnostic repair tools” made available to dealers, on “fair and reasonable terms.” *Mirr*, “Defending the Right to Repair”, 2399. See also H. 4362, 187th Gen. Ct. (Mass. 2012).

[46] Perzanowski, “Consumer Perceptions of the Right to Repair”, 376.

[47] Gabriel Dominic Gomez, "Have Some Heart for the Heartland: A Call for a Federal Right to Repair Law," *Journal of Legislation* 50, no. 1 (January 2024): 149–178.

[48] Elizabeth Chamberlain, "Right to Repair Laws Have Now Been Introduced in All 50 US States," iFixit, February 24, 2025, www.ifixit.com/News/108371/right-to-repair-laws-have-now-been-introduced-in-all-50-us-states.

[49] Wyoming H.B. 0199, 2017 Legislature, General Session (2017).

[50] These bills define "digital electronic products" as "a part or product containing a microprocessor originally manufactured for distribution and sale in the United States," which would ostensibly include everything from a smartphone or computer to household appliances and cars, trucks, or tractors. Iowa H.F. 556, 87th General Assembly, Regular Session, § 1(3) (2017).

[51] *Mirr*, “Defending the Right to Repair”, 2402-2403.

[52] *Id.*, 2405. According to Senator Brasch, who introduced Nebraska's bill, Apple sent its representative Steve Kester to speak with her. Kester reportedly told Senator Brasch that if Nebraska enacted a Right to Repair law, it would become a “mecca” for bad actors and hackers. These technology and equipment companies were joined by groups like the Association of Equipment Manufacturers (“AEM”) and Equipment Dealers Association (“EDA”) who jointly authored a “Statement of Principles” which frames the “right to repair” movement narrowly and not as a “right to modify” movement. Their website claimed that the right to repair laws proposed in state legislatures “would jeopardize the safety and sustainability regulations governing

modern farm equipment." The website jointly operated by AEM and EDA claimed that, in their current form, the right to repair laws jeopardize: (1) safety, as users would be able to override safety features; (2) sustainability, as emissions standards could be circumvented; and (3) innovation, as permitting "access to source code would jeopardize manufacturers' intellectual property and stifle innovation." This website, at one point accessible under the domain R2RSolutions.org, has since been deactivated. The Fair Repair Act was indefinitely postponed in 2018.

[53] Callum Borchers, "Mass. Voters Say 'Yes' on Question 1, Expanding Access to Car Repair Data," WBUR News, November 3, 2020, www.wbur.org/news/2020/11/03/ballot-question-1-right-to-repair-pas-ses. With the passage of Question 1, automakers would be required to make mechanical data available to drivers through a mobile app and, with drivers' consent, to independent shops through a new database. This ballot measure was passed despite fierce opposition from automakers, who spent upwards of \$26 million in lobbying in ad campaigns alleging the measure as a cybersecurity risk.

[54] Washington H.B. 1483, Regular Session (2025); see also Washington S.B. 5680, Regular Session (2025). H.B. 1483 supports the servicing and repair of certain products with digital electronics in a secure and reliable manner to increase access and affordability for Washingtonians, while S.B. 5680 established a right to repair for mobility equipment (i.e., scooters and wheelchairs) for people with physical disabilities.

[55] "Our Members," Alliance for Automotive Innovation, accessed November 14, 2025, www.autosinnovate.org/about/our-members.

[56] *Memorandum of Decision*, Alliance for Automotive Innovation v. Andrea Joy Campbell, Attorney General of the Commonwealth of Massachusetts, United States District Court for the District of Massachusetts, February 11, 2025. AAI contended that the requirements of the Massachusetts law conflicted with the "make inoperative" provision of the MVSA, which prohibits OEMs from "knowingly [making] inoperative any part of a device or element of

design installed on or in a motor vehicle or motor vehicle equipment in compliance with an applicable motor vehicle safety standard." It further asserted that the Massachusetts law was preempted by various federal safety standards that concern acceleration, braking, steering and air bag systems because OEMs "have installed a variety of cybersecurity protections around regulated vehicle functions" and "[y]et the [Massachusetts law] requires motor vehicle manufacturers to remove cybersecurity protections." District Court Judge Denise Casper was unconvinced, holding: "AAI has not met its burden to prove that the [Massachusetts law] stands as an obstacle to the accomplishment and execution of the full purposes and objectives of Congress with respect to the MVSA... No provision in the MVSA directly addresses motor vehicle cybersecurity or access to motor vehicle diagnostic data, nor do any of the [safety standards] that AAI cites discuss cybersecurity protections, much less adopt a federal cybersecurity objective that conflicts with the [Massachusetts law]."

[57] "Class Action Complaint," *Forest River Farms v. Deere & Co.*, United States District Court, Northern District of Illinois, Eastern Division, January 12, 2022.

[58] *Id.*

[59] Tom Koch and Stephanie T. Nguyen, "FTC, States Sue Deere & Company to Protect Farmers from Unfair Corporate Tactics, High Repair Costs," Federal Trade Commission, January 15, 2025, www.ftc.gov/news-events/news/press-releases/2025/01/ftc-states-sue-deere-company-protect-farmers-unfair-corporate-tactics-high-repair-costs. See also "Complaint," *Federal Trade Commission, State of Illinois, and State of Minnesota v. Deere & Co.*, United States District Court for the Northern District of Illinois, Western Division, Case No. 3:25-cv-50017. The FTC alleged that, for decades, Deere's unlawful practices have limited the ability of farmers and independent repair providers to repair Deere equipment, forcing farmers to instead rely on Deere's network of authorized dealers for necessary repairs, and that this unfair steering practice has boosted Deere's multi-billion dollar profits on agricultural equipment and parts, growing its repair parts

business while burdening farmers with higher repair costs. In the words of then-FTC chair, Lina Khan: "The FTC's action today seeks to ensure that farmers across America are free to repair their own equipment or use repair shops of their choice—lowering costs, preventing ruinous delays, and promoting fair competition for independent repair shops."

[65] Rod Sutton, "Deere Motion to Drop Right to Repair Lawsuit Denied by Court," *Construction Equipment*, December 1, 2023, www.constructionequipment.com/equipment-maintenance-and-repair/news/33015818/deere-motion-to-drop-right-to-repair-lawsuit-denied-by-court. See also Todd Neeley, "Federal Court Rejects Deere's Motion to Dismiss FTC Right-to-Repair Lawsuit," *DTN Progressive Farmer*, June 11, 2025, www.dtnpf.com/agriculture/web/ag/equipment/article/2025/06/11/federal-court-rejects-deeres-motion. In the case of the class action, Deere argued that the lawsuit should be dismissed because farmer groups bringing the suit lacked antitrust standing and failed to "plausibly allege relevant markets." District Court Judge Iain Johnston disagreed, holding in a November 2023 ruling that standing does exist and that the plaintiffs had indeed alleged constitutional and antitrust standing, relevant markets, and the necessary requirements in the complaint. The story seems to repeat itself verbatim in 2025 with the FTC action. Deere filed a motion to dismiss, arguing that the statute of limitations had run out on the states' claims and that the states lacked standing. The company also again argued that the plaintiffs failed to plead a cognizable aftermarket, and that farmers had full knowledge of Deere's decades-long repair practices when they bought equipment. Judge Johnston again disagreed and denied the motion to dismiss, this time stating, "Sequels so rarely beat their originals that even the acclaimed Steve Martin couldn't do it on three tries."

[61] "Rulemaking Proceedings under Section 1201 of Title 17," U.S. Copyright Office, accessed November 14, 2025, www.copyright.gov/1201/.

[62] Mitch Stoltz et al., "2015 DMCA Rulemaking," Electronic Frontier Foundation, December 14, 2020, www.eff.org/tl/cases/2015-dmca-rulemaking.

[63] U.S. Copyright Office, "Rules and Regulations," *Federal Register* 89, no. 208 (October 28, 2024), www.govinfo.gov/content/pkg/FR-2024-10-28/pdf/2024-24563.pdf.

[64] Federal Trade Commission, "FTC and DOJ File Comment with the U.S. Copyright Office Supporting Renewal and Expansion of Exemptions Facilitating Consumers' and Businesses' Right to Repair Their Own Products," March 14, 2024, www.ftc.gov/news-events/news/press-releases/2024/03/ftc-doj-file-comment-us-copyright-office-supporting-renewal-expansion-exemptions-facilitating.

[65] *Brown v. Board of Education of Topeka* (1), 347 U.S. 483 (1954)

[66] *Griswold v. Connecticut*, 381 U.S. 479 (1965)

[67] *Obergefell v. Hodges*, 576 U.S. __ (2015)

[68] *Quanta Computer, Inc.*, 553 U.S. 617 (2008), 628.

[69] "About." *PIRG*, 19 Aug. 2025, www.pirg.org/about/. "Being a consumer advocate means being a champion for a marketplace that's more honest and transparent, and a society that's healthier, safer and more sustainable. It means advocating less waste in favor of reducing, reusing and recycling more. It means preventing toxic threats to our health and safety. And it means considering the impact of new products and technologies on the next generation, not just the next quarterly earnings report."

[70] *In re Search of a Residence in Oakland*, 354 F. Supp. 3d 1010 (N.D. Cal. 2019), 1. "The Government, however, also seeks the authority to compel any individual present at the time of the search to press a finger (including a thumb) or utilize other biometric features, such as facial or iris recognition, for the purposes of unlocking the digital devices found in order to permit a search of the contents as authorized by the search warrant. For the reasons set forth below, the Court finds that the Government's request runs afoul of the Fourth and Fifth Amendments, and the search warrant application must be DENIED."

[71] *Id.* at 3-4. "The Court finds that utilizing a biometric feature to unlock an electronic device is not akin to submitting to fingerprinting or

a DNA swab, because it differs in two fundamental ways. First, the Government concedes that a finger, thumb, or other biometric feature may be used to unlock a device in lieu of a passcode. In this context, biometric features serve the same purpose of a passcode, which is to secure the owner's content, pragmatically rendering them functionally equivalent.”

[72] *Id.* at 3.

[73] Mirr, “Defending the Right to Repair”, 2397.

[74] “Software License agreement for MacOS Sequoia”, Apple Inc., July 11, 2024. <https://www.apple.com/legal/sla/docs/macOSSequoia.pdf>.

[75] “DE’s Market Share Relative to Its Competitors, as of Q1 2026.” CSIMarket, June 16, 2026. <https://csimarket.com/stocks/competition/SEG2.php?code=DE>.

[76] “Lenovo License Agreement (COE-30002-01)”, Lenovo Support US, July 2022. <https://support.lenovo.com/us/en/solutions/ht100141-lenovo-license-agreement-l505-0009-06>. See also “End-User License Agreement.” HP, June 2021. https://support.hp.com/us-en/document/ish_4416646-4390016-16.

[77] *Quanta Computer, Inc.*, 553 U.S. 617, 629 (2008), 1. “A separate agreement (Master Agreement) required Intel to give its customers written notice that the license does not extend to a product made by combining an Intel Product with a non-Intel product, and provided that a breach of the agreement would not affect the License Agreement. Petitioner computer manufacturers (Quanta) purchased microprocessors and chipsets from Intel. Quanta then manufactured computers using Intel parts in combination with non-Intel parts, but did not modify the Intel components. LGE sued, asserting that this combination infringed the LGE Patents.”

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